LIFTGATE GLASS/WINDSHIELD GLASS REPLACEMENT

AFFECTED VEHICLES: CERTAIN 2000 MODEL YEAR WINDSTAR

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OVERVIEW

This procedure is for replacement of the liftgate glass, windshield glass, and urethane adhesive/sealer. All vehicles affected by this program will have the liftgate glass replaced. Some of the vehicles will also have the windshield replaced. One urethane installation kit is required for each piece of glass.

Be sure you have the appropriate quantity of materials on hand before scheduling the customer service appointment. Please refer to Attachment II of Recall 00S01 for ordering instructions.

NOTE: **<u>Use only</u> Essex product #59202, Urethane Installation Kit, when performing this procedure. If incorrect urethane is used, the glass will not bond properly to the body flange opening.**

PREPARATION

1. NOTE: Installation kits are labeled with an expiration date. Expired kits must be reordered.

   Verify installation kit(s) are within the expiration date.

2. **CAUTION: Open the windows to prevent the *newly* installed glass from being pushed out by air pressure if a door is closed. The windows must remain open throughout the 24-hour urethane curing period.**

   Open the vehicle windows.

3. Protect the vehicle from damage by covering the rear cargo carpeting. Also, if the windshield glass is being replaced, cover the front seats, instrument panel and hood.

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GLASS REMOVAL

WARNING: TO PREVENT GLASS SPLINTERS FROM ENTERING EYES OR CUTTING HANDS, WEAR SAFETY GLASSES AND HEAVY GLOVES WHEN CUTTING GLASS FROM VEHICLE.

LIFTGATE (all vehicles)

1. Remove the rear wiper arm.
2. Remove the liftgate interior trim panel.
3. Disconnect and remove the rear wiper motor.
4. Disconnect and remove the high-mounted stoplamp.
5. Disconnect the rear defroster.
6. **CAUTION: Protective tape must be applied to the liftgate to avoid damaging the painted surface.**

   Clean the liftgate around the glass opening. Then apply protective tape to the painted surface around the glass opening. See Figure 1.

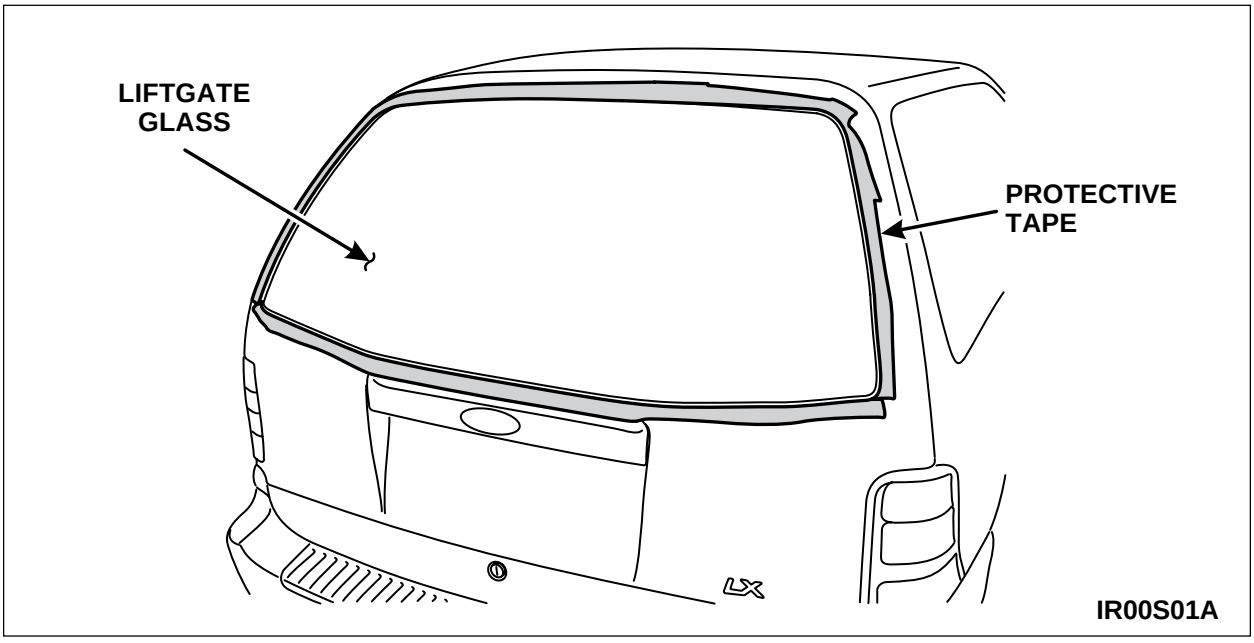


FIGURE 1

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7. NOTE: Leave as much of the urethane on the liftgate body as possible to aid in peeling the urethane from the body.

   Using a suitable glass removal cold knife with a 1.5" or 2" blade, cut the urethane from the liftgate glass, then remove the glass. See Figure 2.

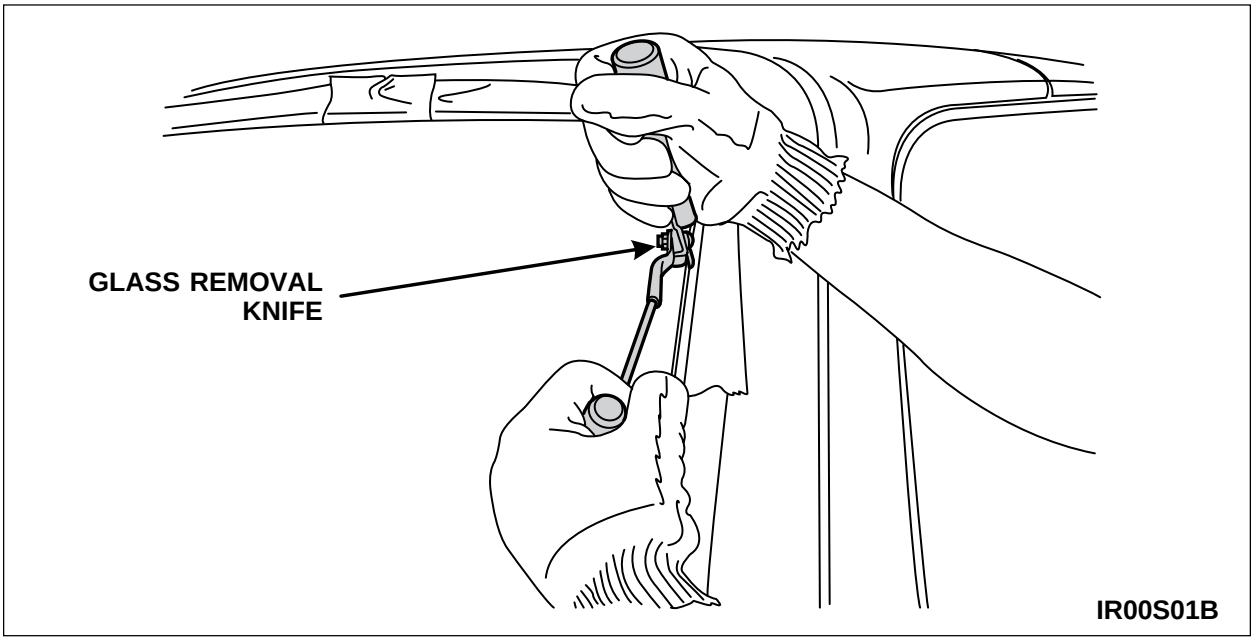


FIGURE 2

8. Remove the protective tape from the liftgate.
9. Remove the rear wiper motor shaft grommet from the liftgate glass.
10. Proceed to **Urethane Adhesion Test** (Page 5).

WINDSHIELD (some vehicles)

1. Remove the interior rearview mirror.
2. Remove the left and right interior A-pillar mouldings.
3. Remove the instrument panel defroster-opening grille. Disconnect the sunload sensor, if equipped.
4. Remove the windshield wiper arms.
5. Remove the cowl top vent panel.

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6. Cut off the rubber portion of the windshield reveal moulding with a utility knife. See Figure 3.

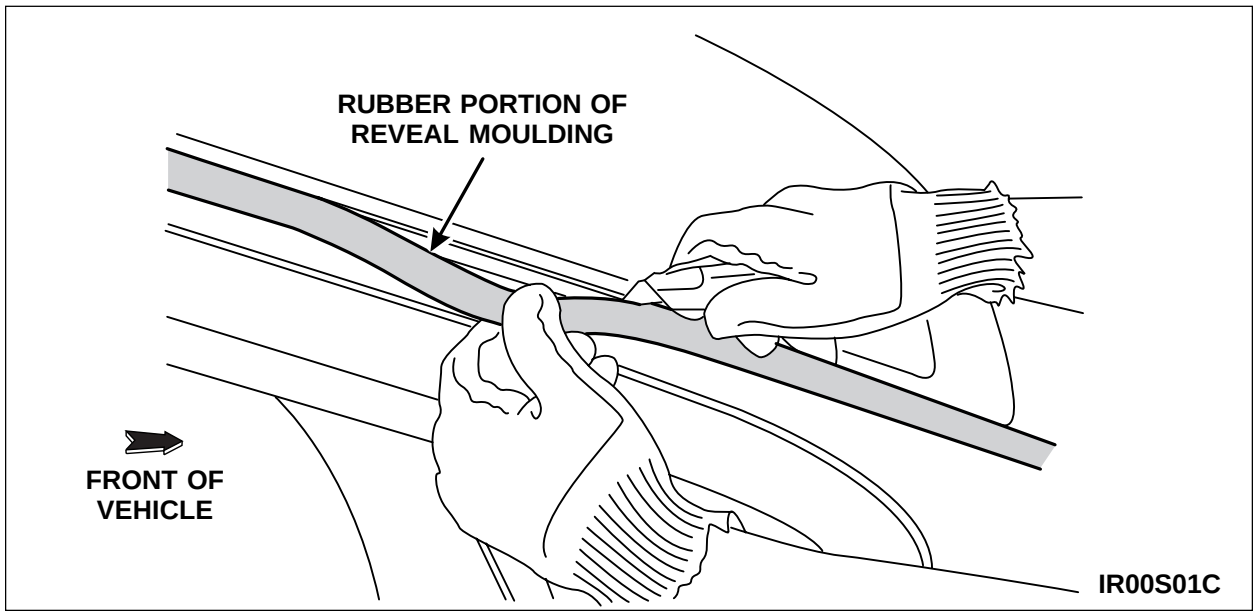


FIGURE 3

7. NOTE: Leave as much of the urethane on the vehicle body as possible, to aid in peeling the urethane from the body.

   Using a suitable glass removal cold knife with a 1.5" or 2" blade, cut the urethane from the windshield glass, then remove the glass. See Figure 4.

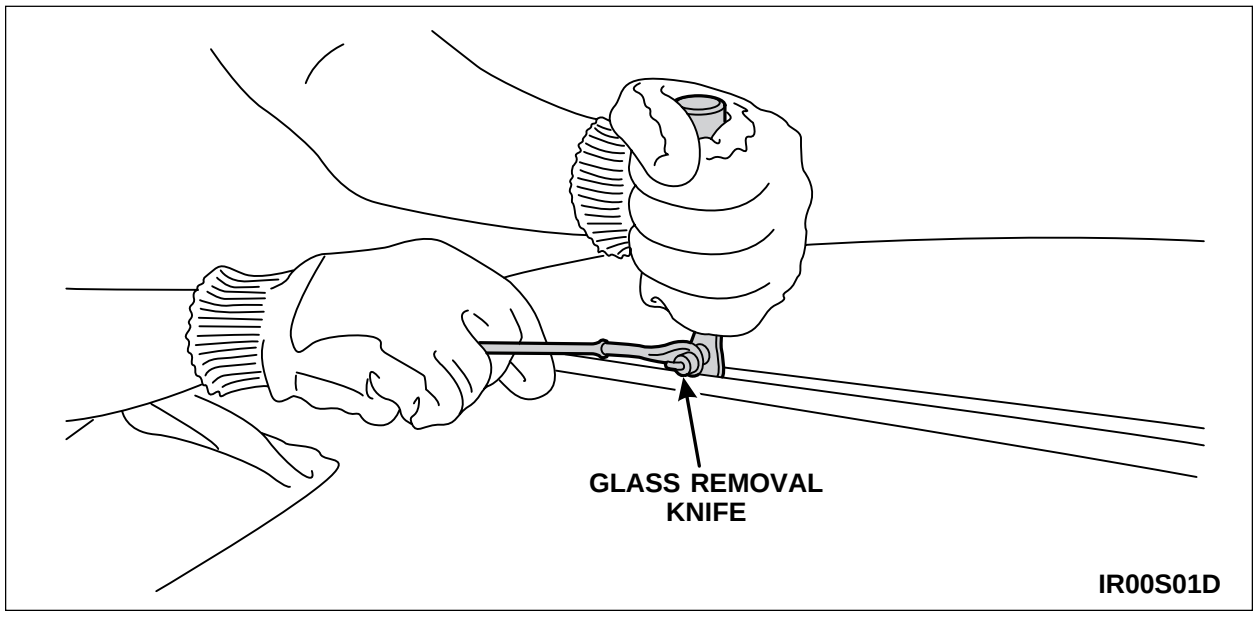


FIGURE 4

8. Proceed to **Urethane Adhesion Test** (Page 5).

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URETHANE ADHESION TEST

NOTE: Be sure to perform this procedure on the liftgate glass (all affected vehicles) and on the windshield glass (some vehicles).

NOTE: The purpose of this procedure is to locate and remove all urethane that is not bonded properly to the body flange opening. In some areas, the urethane will be bonded properly and in other areas the urethane will peel from the painted body flange opening when tested. Be sure to completely remove all urethane that is not bonded properly to the body flange opening.

1. **CAUTION: Do not cut into the painted surface of the window opening flange when removing the urethane. If the painted surface is scratched or cut, be sure to coat with body primer contained in the Urethane Installation Kit Essex #59202.**

   At the top window opening flange, cut into the urethane at an angle to form a tab. Attempt to peel the urethane from the flange by pulling up on the urethane tab.
   - Remove all urethane that can be peeled from the flange.
   - Trim all remaining urethane that is firmly bonded to the flange to a thickness of 1-2 mm. See Figures 5 and 6.

2. Repeat the previous step for the remaining three sides of the window opening.

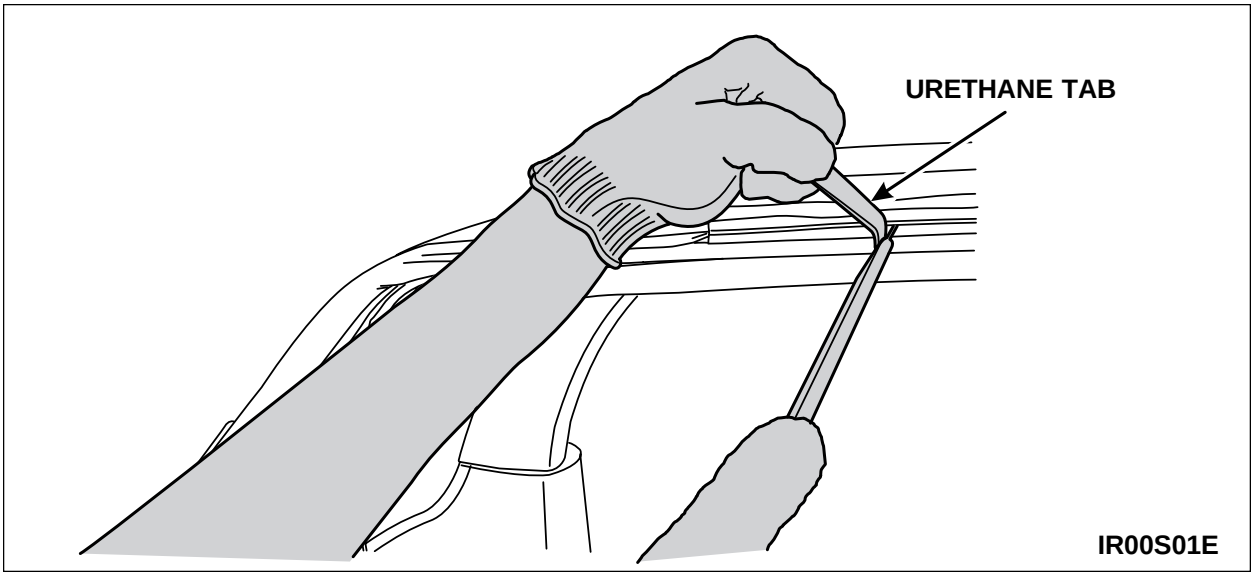


FIGURE 5

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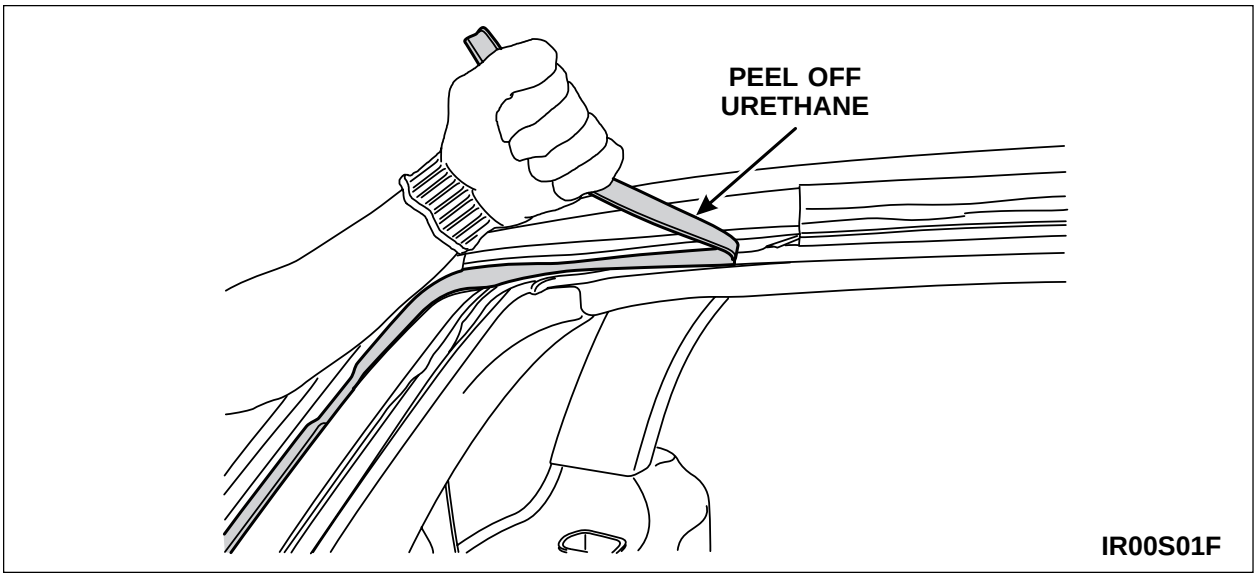


FIGURE 6

ADHESIVE PREPARATION AND APPLICATION

WARNING: RUBBER GLOVES AND SAFETY GLASSES MUST BE WORN WHEN HANDLING PRIMERS. CAREFULLY OPEN THE BOTTLES AND INSERT CLEAN, UNUSED DAUBERS INTO THE BOTTLES. TO AVOID SPILLING, NEVER POUR PRIMERS ONTO THE DAUBERS. REPLACE INNER SEAL AND CAP ON BOTTLES IMMEDIATELY AFTER USE.

PREPARATION

1. Remove all dirt and debris from the window opening flange.
2. Clean the painted bonding surface and/or any remaining urethane on the window opening flange with either:
   - Naphtha, or
   - an alcohol solution that contains a minimum of 90% alcohol.
3. With the aid of an assistant, dry fit the liftgate/windshield glass to the window opening. Mark this position with masking tape as guides for installation.

APPLICATION

BODY PRIMER

NOTE: Only chipped or scratched areas will be primed. If no painted areas are damaged, body priming is not required.

1. Inspect the window opening flange for any chipped paint or scratches.
2. Shake the bottle of Body Primer for at least one minute prior to application.
3. Apply Body Primer to the chipped or scratched painted areas of the body flange. Avoid priming any remaining urethane.
4. Allow Body Primer to dry for 30 minutes.

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GLASS PREP #1

1. Wipe glass clean with a lint-free cloth.
2. Shake the bottle of Glass Prep #1 for at least one minute prior to application.
3. Apply Glass Prep #1 to one side of the perimeter (bond line where the urethane contacts the glass) of the glass. Then immediately wipe with a clean, dry, lint-free cloth. Repeat this process on the remaining three sides of the perimeter of the glass.
4. Repeat step 3 on the entire perimeter of the glass, for a total of two applications of Glass Prep #1.
5. Allow Glass Prep #1 to dry for at least 30 seconds, but no more than 10 minutes before applying Glass Primer #2.

GLASS PRIMER #2

1. Shake the bottle of glass primer #2 for at least one minute prior to application.
2. Apply glass primer #2 directly on top of Glass Prep #1 to the entire perimeter of the glass.
3. Allow glass primer #2 to dry at least one minute.

URETHANE ADHESIVE

CAUTION: Immediately remove any excess urethane adhesive from the painted body surface of the vehicle.

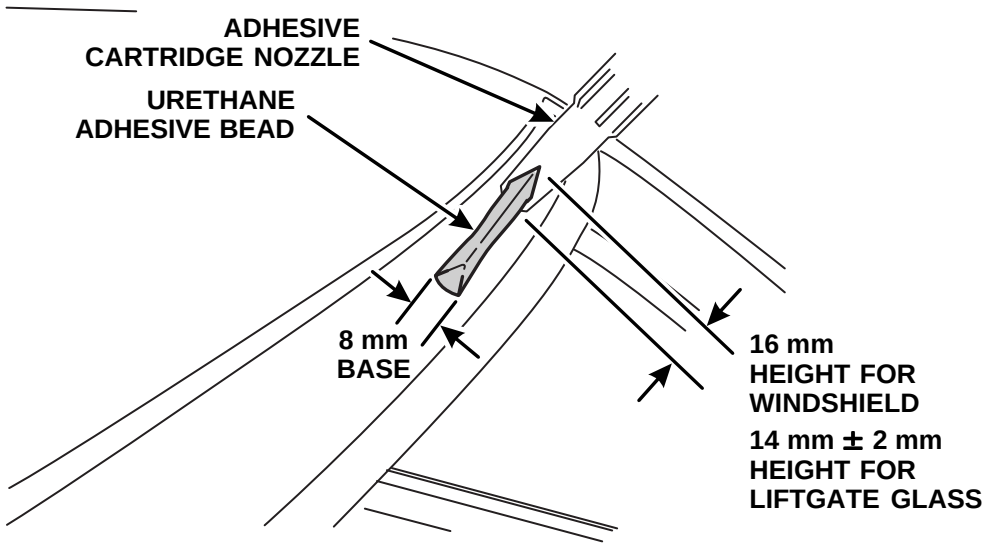
NOTE: Do not use pneumatic-powered guns for applying adhesive.

1. Load the urethane adhesive cartridge into a suitable application gun.
2. Cut the urethane adhesive cartridge nozzle in a “V” shape to produce a triangle shaped bead.

NOTE: For the liftgate glass installation, the urethane adhesive is applied directly on the Glass Primer #2 on the perimeter of the glass. For the windshield glass installation, the urethane adhesive is applied to the windshield opening flange. Apply urethane adhesive directly over any existing, properly bonded urethane on the windshield opening flange.

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3. Hold the cartridge nozzle perpendicular to the glass or flange when dispensing the urethane adhesive. Use a smooth, continuous motion to form a uniform triangular bead. The recommended size of the bead is 8 mm across the base and 16 mm in height for the windshield, and 14 mm in height for the liftgate. See Figure 7.



WINDSHIELD GLASS OPENING SHOWN
URETHANE ADHESIVE IS APPLIED TO GLASS FOR LIFTGATE PROCEDURE
IR00S01G

FIGURE 7

4. Verify that the bead is uniform and has no gaps. Add material, if necessary.
5. The glass must be installed within 10 minutes after applying the urethane adhesive.
6. Proceed to **Glass Installation, Liftgate** (Page 8) or **Windshield (if required)** (Page 9).

GLASS INSTALLATION

LIFTGATE

1. Install the rear wiper motor shaft grommet onto the liftgate glass.
2. Position the liftgate glass into the glass opening using the previously applied masking tape as a guide. Press the glass into place.
3. Connect the rear defroster.
4. Install and connect high-mounted stoplamp.
5. Install and connect the rear wiper motor. Tighten the bolts to 11 Nm (97 lb-in).
6. Install the liftgate interior trim panel.
7. Install the rear wiper arm. Tighten the nut to 15 Nm (133 lb-in).
8. Leave the liftgate in the full open position for one hour.

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WINDSHIELD (if required)

1. Install the interior rearview mirror onto the windshield glass.
2. Install the instrument panel defroster opening grille. Connect the sunload sensor, if equipped.
3. Install the left and right interior A-pillar mouldings.
4. With the aid of an assistant, position the windshield glass into the window opening using the masking tape guides. Press the glass into place.
5. Install the cowl top vent panel.
6. Install the windshield wiper arms. Tighten the nuts to 35 Nm (26 lb-ft).

CLEANUP AND CURING

1. Remove all the protective covers that were installed.
2. **WARNING: URETHANE ADHESIVE MUST CURE FOR AT LEAST 24 HOURS AT A MINIMUM OF 21ºC (70ºF) BEFORE DELIVERING THE VEHICLE TO THE CUSTOMER.**

   Allow the urethane to cure for 24 hours, then clean the ***newly*** installed glass.
3. Do not save any unused material from the kit(s). Properly dispose of all unused material.

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